



ULTIMATE JS CHEATSHEET

by- CreativeSiddharth

Basics

Including JavaScript in an HTML Page

```
<script type="text/javascript">  
  JS code goes here  
</script>
```

Call an External JavaScript File

```
<script src="myscript.js"></script>
```

Including Comments

```
//Single line comments  
/* Multi-line comments */
```

Variables

VAR, CONST, LET

var

Can be reassigned but only accessed within a function.

const

Cannot be reassigned and not accessible before they appear within the code

let

Similar to const, however, it can be reassigned but not re-declared

DATA TYPES

Numbers

```
var age = 20
```

Text (strings)

```
var a = "siddharth"
```

Boolean

```
var b = true
```

Constant numbers

```
const PI = 3.14
```

Objects

```
var name = {  
  firstName: "siddharth",  
  lastName: "pandey"  
}
```

Arrays

```
var fruit = ["Banana", "Apple", "Pear"];
```

ARRAY METHODS

concat()

Join several arrays into one

indexOf()

Returns the first position at which a given element appears in an array

join()

Combine elements of an array into a single string and return the string

lastIndexOf()

Gives the last position at which a given element appears in an array

pop()

Removes the last element of an array

push()

Add a new element at the end

reverse()

Reverse the order of the elements in an array

shift()

Remove the first element of an array

slice()

Pulls a copy of a portion of an array into a new array

splice()

Adds elements in a specified way and position

toString()

Converts elements to strings

unshift()

Adds a new element to the beginning

valueOf()

Returns the primitive value of the specified object

Operators

Basic Operators

+ Addition

- Subtraction

* Multiplication

/ Division

(..) Grouping operator

% Modulus (remainder)

++ Increment numbers

-- Decrement numbers

Bitwise Operators

& AND statement

| OR statement

~ NOT

^ XOR

<< Left shift

>> Right shift

>>> Zero fill right shift

Logical Operators

&& Logical and

|| Logical or

! Logical not

Comparison Operators

== Equal to

=== Equal value and equal type

!= Not equal

!== Not equal value or not equal type

> Greater than

< Less than

>= Greater than or equal to

<= Less than or equal to

? Ternary operator

Functions

```
function name(parameter1, parameter2, ..) {  
    what the function does  
}
```

OUTPUTTING DATA

alert()

Output data in an alert box in the browser window

confirm()

Opens up a yes/no dialog and returns true/false depending on user click

console.log()

Writes information to the browser console, good for debugging purposes

document.write()

Write directly to the HTML document

prompt()

Creates an dialogue for user input

GLOBAL FUNCTIONS

decodeURI()

Decodes a Uniform Resource Identifier (URI) created by encodeURI or similar

decodeURIComponent()

Decodes a URI component

encodeURIComponent()

Encodes a URI into UTF-8

encodeURIComponentComponent()

Same but for URI components

eval()

Evaluates JavaScript code represented as a string

isFinite()

Determines whether a passed value is a finite number

isNaN()

Determines whether a value is NaN or not

Number()

Returns a number converted from its argument

parseFloat()

Parses an argument and returns a floating point number

parseInt()

Parses its argument and returns an integer

Loops

for()

```
for (before loop; condition for loop; execute after loop) {  
    write what to do during the loop  
}
```

while()

```
initialization  
while(condition for loop){  
    write what to do during the loop  
    reinitialization  
}
```

do while()

```
initialization  
do(){  
    write what to do during the loop  
    reinitialization  
}while(condition for loop)
```


if-else

```
if (condition) {  
  // what to do if condition is met  
} else {  
  // what to do if condition is not met  
}
```

String

```
var person = "Siddharth Pandey";
```

STRING METHODS

charAt()

Returns a character at a specified position

charCodeAt()

Gives you the unicode of character at that position

concat()

Concatenates (joins) two or more strings into one

indexOf()

Provides the position of the first occurrence of a specified text within a string

lastIndexOf()

Same as indexOf() but with the last occurrence

match()

Retrieves the matches of a string against a search pattern

replace()

Find and replace specific text in a string

search()

Executes a search for a matching text and returns its position

slice()

Same as indexOf() but with the last occurrence

split()

Splits a string object into an array of strings at a specified position

substr()

Similar to slice() but extracts a substring depended on a specified number of characters

toLowerCase()

Convert strings to lowercase

toUpperCase()

Convert strings to uppercase

valueOf()

Returns the primitive value (that has no properties or methods) of a string object

Numbers and Math

NUMBER PROPERTIES

MAX_VALUE

The maximum numeric value representable

MIN_VALUE

Smallest positive numeric value representable

NaN

The “Not-a-Number” value

NEGATIVE_INFINITY

Smallest positive numeric value representable

POSITIVE_INFINITY

Positive Infinity value

NUMBER METHODS

toExponential()

Returns a string with a rounded number written as exponential notation

toFixed()

Returns the string of a number with a specified number of decimals

toPrecision()

String of a number written with a specified length

toString()

Returns a number as a string

MATH PROPERTIES

E

Euler's number

LN2

The natural logarithm of 2

LN10

Natural logarithm of 10

LOG2E

Base 2 logarithm of E

PI

number PI

SQRT1_2

Square root of 1/2

SQRT2

The square root of 2

MATH METHODS

abs(x)

Returns the absolute (positive) value of x

acos(x)

The arccosine of x, in radians

asin(x)

Arcsine of x, in radians

ceil(x)

Value of x rounded up to its nearest integer

cos(x)

The cosine of x (x is in radians)

floor(x)

The value of x rounded down to its nearest integer

log(x)

The natural logarithm (base E) of x

max(x,y,z,...,n)

Returns the number with the highest value

min(x,y,z,...,n)

Same for the number with the lowest value

pow(x,y)

X to the power of y

random()

The value of x rounded to its nearest integer

sin(x)

The sine of x (x is in radians)

sqrt(x)

Square root of x

tan(x)

The tangent of an angle

Dates

SETTING DATES

Date()

Creates a new date object with the current date and time

Date(2021, 5, 21, 3, 23, 10, 0)

Create a custom date object

Date("202-06-23")

Date declaration as a string

PULLING DATE AND TIME VALUES

getDate()

Get the day of the month as a number (1-31)

getDay()

The weekday as a number (0-6)

getFullYear()

Year as a four digit number (yyyy)

getHours()

Get the hour (0-23)

getMinutes()

Get the minute (0-59)

getMonth()

Month as a number (0-11)

getSeconds()

Get the second (0-59)

getTime()

Get the milliseconds since January 1, 1970

SET PART OF A DATE

setDate()

Set the day as a number (1-31)

setFullYear()

Sets the year (optionally month and day)

setHours()

Set the hour (0-23)

setMinutes()

Sets the minutes (0-59)

setMonth()

Set the month (0-11)

setSeconds()

Sets the seconds (0-59)

setTime()

Set the time (milliseconds since January 1, 1970)

DOM Node

NODE PROPERTIES

attributes

Returns a live collection of all attributes registered to and element

baseURI

Provides the absolute base URL of an HTML element

childNodes

Gives a collection of an element's child nodes

firstChild

Returns the first child node of an element

lastChild

The last child node of an element

nextSibling

Gives you the next node at the same node tree level

nodeName

Returns the name of a node

nodeType

Returns the type of a node

nodeValue

Sets or returns the value of a node

ownerDocument

The top-level document object for this node

parentNode

Returns the parent node of an element

previousSibling

Returns node immediately preceding the current one

textContent

Sets or returns the textual content of a node and its descendants

NODE METHODS

appendChild()

Adds a new child node to an element as the last child node

cloneNode()

Clones an HTML element

hasAttributes()

Returns true if an element has any attributes else false

hasChildNodes()

Returns true if an element has any child nodes, else false

insertBefore()

Inserts new child node before specified, existing child node

isEqualNode()

Checks if two elements are equal

isSameNode()

Checks if two elements are the same node

isSupported()

Returns true if a specified feature is supported

removeChild()

Removes a child node from an element

replaceChild()

Replaces a child node in an element

ELEMENT METHODS

getAttributeNS()

Returns string value of the attribute with the specified namespace and name

getAttribute()

Returns the specified attribute value of an element node

getAttributeNode()

Gets the specified attribute node

getElementsByTagName()

Provides a collection of all child elements with the specified tag name

hasAttribute()

Returns true if an element has any attributes, else false

removeAttribute()

Removes a specified attribute from an element

removeAttributeNode()

Takes away a specified attribute node and returns the removed node

setAttribute()

Sets or changes specified attribute to specified value

setAttributeNode()

Sets or changes the specified attribute node

setAttributeNodeNS()

Adds a new namespaced attribute node to an element

Working with Browser

WINDOW PROPERTIES

closed

Checks whether a window has been closed or not and returns true or false

defaultStatus

Sets or returns default text in the statusbar of window

document

Returns the document object for the window

history

Provides the History object for the window

innerHeight

The inner height of a window's content area

innerWidth

The inner width of the content area

location

Returns the location object for the window

name

Sets or returns the name of a window

navigator

Returns the Navigator object for the window

opener

Returns a reference to the window that created the window

outerHeight

outer height of a window, including scrollbars/Toolbar

outerWidth

outer width of a window same as outerHeight

pageXOffset

Number of pixels the current document has been scrolled horizontally

pageYOffset

Number of pixels the document has been scrolled vertically

parent

The parent window of the current window

screen

Returns the Screen object for the window

screenLeft

The horizontal coordinate of the window

screenTop

The vertical coordinate of the window

screenX

Same as screenLeft but needed for some browsers

screenY

Same as screenTop but needed for some browsers

status

Sets or returns the text in the statusbar of a window

WINDOW METHODS

alert()

Displays an alert box with a message and an OK button

blur()

Removes focus from the current window

clearInterval()

Clears a timer set with setInterval()

alert()

Displays an alert box with a message and an OK button

clearTimeout()

Clears a timer set with setTimeout()

close()

Closes the current window

confirm()

Displays a dialogue box with a message and an OK and Cancel button

focus()

Sets focus to the current window

moveBy()

Moves a window relative to its current position

moveTo()

Moves a window to a specified position

open()

Opens a new browser window

print()

Prints the content of the current window

prompt()

Displays a dialogue box that prompts the visitor for input

resizeBy()

Resizes the window by the specified number of pixels

resizeTo()

Resizes the window to a specified width and height

scrollBy()

Scrolls the document by a specified number of pixels

scrollTo()

Scrolls the document to specific coordinates

setInterval()

Calls a function or evaluates an expression at specified intervals

setTimeout()

Calls a function or evaluates an expression after a specified interval

stop()

Stops the window from loading

SCREEN PROPERTIES

availHeight / availWidth

Returns the height / width of the screen (excluding the Windows Taskbar)

height / width

The total height / width of the screen

colorDepth

Returns the bit depth of the color palette for displaying images

pixelDepth

The color resolution of the screen in bits per pixel

Events

MOUSE

onclick

The event occurs when the user clicks on an element

oncontextmenu

User right-clicks on an element to open a context menu

ondblclick

The user double-clicks on an element

onmousedown

User presses a mouse button over an element

onmouseenter

The pointer moves onto an element

onmouseleave

Pointer moves out of an element

onmousemove

The pointer is moving while it is over an element

onmouseover

When the pointer is moved onto an element

onmouseout

User moves the mouse pointer out of an element

KEYBOARD

onkeydown

When the user is pressing a key down

onkeypress

The moment the user starts pressing a key

onkeyup

The user releases a key

FRAME

onabort

The loading of a media is aborted

onbeforeunload

occurs before the document is about to be unloaded

onerror

An error occurs while loading an external file

onhashchange

There have been changes to the anchor part of a URL

onload

When an object has loaded

onpagehide

The user navigates away from a webpage

onpageshow

When the user navigates to a webpage

onresize

The document view is resized

onscroll

An element's scrollbar is being scrolled

onunload

Event occurs when a page has unloaded

FORM

onblur

When an element loses focus

onchange

The content of a form element changes
(for <input>, <select> and <textarea>)

onfocus

An element gets focus

onfocusin

When an element is about to get focus

onfocusout

The element is about to lose focus

oninput

User input on an element

oninvalid

An element is invalid

onreset

A form is reset

onsearch

The user selects some text (for `<input>` and `<textarea>`)

onsubmit

A form is submitted

DRAG

ondrag

An element is dragged

ondragend

The user has finished dragging the element

ondragenter

The dragged element enters a drop target

ondragleave

A dragged element leaves the drop target

ondragover

The dragged element is on top of the drop target

ondragstart

User starts to drag an element

ondrop

Dragged element is dropped on the drop target

CLIPBOARD

oncopy

User copies the content of an element

oncut

The user cuts an element's content

onpaste

A user pastes content in an element

ANIMATION

animationend

A CSS animation is complete

animationiteration

CSS animation is repeated

animationstart

CSS animation has started

OTHERS

transitionend

Fired when a CSS transition has completed

onmessage

A message is received through the event source

onoffline

Browser starts to work offline

ononline

The browser starts to work online

onpopstate

When the window's history changes

onshow

A <menu> element is shown as a context menu

onstorage

A Web Storage area is updated

ontoggle

The user opens or closes the <details> element

onwheel

Mouse wheel rolls up or down over an element

ontouchcancel

Screen touch is interrupted

ontouchmove

A finger is dragged across the screen

ontouchstart

Finger is placed on touch screen

Error

try

Lets you define a block of code to test for errors

catch

Set up a block of code to execute in case of an error

throw

Create custom error messages instead of the standard JavaScript errors

finally

Lets you execute code, after try and catch, regardless of the result

ERROR NAME AND VALUES

name

Sets or returns the error name

message

Sets or returns an error message in string form

EvalError

An error has occurred in the eval() function

RangeError

A number is "out of range"

ReferenceError

An illegal reference has occurred

SyntaxError

A syntax error has occurred

TypeError

A type error has occurred

URIError

An encodeURI() error has occurred

Regular Expressions

PATTERN MODIFIERS

- e** — Evaluate replacement
- i** — Perform case-insensitive matching
- g** — Perform global matching
- m** — Perform multiple line matching
- s** — Treat strings as single line
- x** — Allow comments and whitespace in pattern
- U** — Non Greedy pattern

BRACKETS

- [abc]** Find any of characters between the brackets
- [^abc]** Find any character not in the brackets
- [0-9]** Used to find any digit from 0 to 9
- [A-z]** Find any character from uppercase A to lowercase z
- (a|b|c)** Find any of the alternatives separated with |

METACHARACTERS

- .** — Find a single character, except newline or line terminator
- \w** — Word character

- `\W` — Non-word character
- `\d` — A digit
- `\D` — A non-digit character
- `\s` — Whitespace character
- `\S` — Non-whitespace character
- `\b` — Find a match at the beginning/end of a word
- `\B` — A match not at the beginning/end of a word
- `\0` — NUL character
- `\n` — A new line character
- `\f` — Form feed character
- `\r` — Carriage return character
- `\t` — Tab character
- `\v` — Vertical tab character
- `\xxx` — The character specified by an octal number xxx
- `\xdd` — Character specified by a hexadecimal number dd
- `\uxxxx` — The Unicode character specified by a hexadecimal number xxxx

QUANTIFIERS

- n^+ — Matches any string that contains at least one n
- n^* — string that contains 0 or more occurrences of n
- $n^?$ — string that contains 0 or one occurrences of n
- $n\{X\}$ — String that contains a sequence of X n 's
- $n\{X,Y\}$ — Strings that contains sequence of X to Y n 's
- $n\$$ — Any string with n at the end of it
- n — String with n at the beginning of it
- $?=n$ — Any string that is followed by specific string n
- $?!n$ — String that is not followed by a specific string n
- $n\{X,\}$ — Matches any string that contains a sequence of at least X n 's

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